980 SOFTWARE DOCUMENTATION

P/N 944102-9902

APPLICATION	
NEXT ASSY	USED ON
944140	8101

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED

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TEXAS INSTRUMENTS
INCORPORATED
Equipment Group Dallas, Texas

PD, PATCH MEMO, DX980 DS330 DISC, REV. *B

SIZE	CODE IDENT NO	DRAWING NO
A	96214	944102-9902
SCALE	REV. C	SHEET 1 of20

I. Patch Procedure for Standard Operating Systems

This document provides the information necessary to update, by means of a series of patches, the DX980 Operating System previously shipped. These patches are to be incorporated into the operating system by means of the dump and modify disc commands. Users not familiar with these commands should reference Section II.

This patch memo is updated periodically through the addition of the latest patches. The actual information needed to apply the patches is given in Tables 1 & 2 and Figure 1. Therefore this patch memo may include several different series of these tables and figures in chronological order, each bearing the date of the patch update.

For a standard system as received from Texas Instruments the infomation provided in Table 1 and the printout of Figure 1 is all that is needed to implement the patches. Table 1 provides the following infomation about each of the patches: a description of the nature of the fix, the absolute sector address and the word offset within the sector for the patch, the present contents of the word and the patch to be applied to the word. Figure 1 provides a complete system console printout of the patch implementation processs performed at Texas Instruments. The user may reproduce this printout at his installation to implement the patch update process on his system. He need only re-IPL following this procedure in order to insure that the patched system is being utilized in memory.

For a nonstandard, or custom, system the procedure for patching the system consists of first calculating the absolute sector address for the patch and then following the standard procedure for applying the patch to the sector. A description of this procedure for nonstandard patches is given in Section III. The information necessary for calculating the address of the sector to be modified is provided the nonstandard user in Table 2.

Nonstandard operating systems may result in several different ways.

1). The DX980 Source Kit (P/N 944140-0010) customer may have generated a custom DX980 Operating System.

2). The customer may have copied (by means of the General Copy Utility DXCOPY) the load module affected by the patch.

3). The customer may have modified (by means of the Load Module Update Utility LMUPDT) an individual MIP affected by the patch. This would alter the disc address at which the MIP starts and in addition could alter the word offset at which the patch should be applied. This latter case must be handled on an individual basis by the customer and cannot be treated in Section III.

If the DX980 system received from Texas Instruments has been modified in any of the three ways described above, please reference Section III and Table 2 before applying the patches.

II. Use of Dump Disc and Modify Disc Commands

Disc patches may be applied with the aid of the Dump Disc and Modify Disc operator communications commands. First, to display the disc sector to be modified a Dump Disc command is entered on the system console. This command is of the form:

//DD DISNO SSEC ESEC

DISNO is an integer that specifies the disc drive number. Disc number 1 is always the system disc. For others discs, this number may be determined by the List Devices IPL command. The disc number is indicated in the ID column of this display.

SSEC is the starting sector number to be displayed. Sector numbers may be decimal (ddddd $\leq$ 32767) or hexadecimal (>hhhh $\leq$ $7FFF_{16}$), where hexadecimal sector numbers are preceded by >.

ESEC is the ending sector number being requested.

The display sector should be verified to match the listing before the patch is applied.

The patch is applied with the Modify Disc command.

//MD DISNO SSEC WRDNO DATA

WRDNO is the word off-set (either decimal or hexadecimal) into the sector to which the patch is to applied. Word offsets range from 0 to 31_{10} or 0 to $1F_{16}$.

DATA represents one or more decimal or hexadecimal numbers separated by blanks. These will be applied to the disc sector consecutively starting at the word offset WRDNO.

After the patch is made the Dump Disc command should again be used to verify the changes.

III. Patch Procedure for Nonstandard Operating Systems

To install a patch to a nonstandard operating system, the customer uses the standard procedure except the correct disc address to be modified must be calculated. This address can be calculated from the information provided in Table 2 for each of the patches. The formula for calculating the absolute disc address is provided at the bottom of Table 2. The procedure is as follows:

1). Run CATFIL to find the starting track numbers of the load modules listed in Table 2.

2). Convert the starting track address for the particular load module listed for the patch to a starting sector address by multiplying by 58_{16}. Enter this sector address in the blank column in the Table labeled "Load Mod Start Sector LS."

3). The MIP number in Table 2 indicates the MIP, or overlay, to be modified. If MIP=0, the patch is to the root segment itself and steps 4 through 7 can be skipped.

4). Add the relative sector number of the MIP directory words for the particular MIP (Shown in Table 2 under column labeled "Directory Words, Sector Offset DS") to the starting sector address of the load module.

5). Display the sector number calculated in Step 4.

6.) The directory words for the MIP to be patched start at the word number indicated under column labled "Directory Words, Word Offset DW." The first word of the directory words gives the sector number for the start of the MIP relative to the beginning of the load module.

7) Add this relative starting sector address to the starting sector address of the load module. The result is the starting address of the MIP.

8) Add to the starting sector address for the MIP (this is equal to the address of the load module if the MIP number is 0) the sector offset within the MIP that is to be patched. Enter this absolute sector address to be patched in the blank column in the table labeled "Calc Abs Sector AS."

9) Display the sector number calculated in step 8. The contents of the displayed sector should match the display of the equivalent sector on the released DX980 system (see Figure 1).

10) Modify the same words in the sector calculated in step 8 as were modified in the released system. The word(s) to modify start with the word indicated in the column labeled "Word Offset into Sector". The present contents of the word(s) to modify and the patch to be applied are provided in the column labeled "Present" and "Patch", respectively.

11) Repeat steps 2 through 10 for each patch to be applied.

12) Re-IPL the patched operating system to ensure that the modified code is used.

As an example of applying a patch to a nonstandard operating system, the procedure is used to apply one of the patches for the Revision ** DX980 Operating System. The module being modified will be one from REV ** DX980 rather than a nonstandard DX980. The excerpt from Table 2 applicable to this example is reproduced here.

ch	Load Module	Load Mod Start Sector LS	MIP #	Directory Words Sector Offset DS	Directory Words Word Offset DW	Sector Offset Into MIP MS	Calc Abs Sector AS	Word Offset into Sector	Present	Patch
1	SYSLD		>C	1	2	>D		>D	>8102	>2102
								>E	>0700	>8102
								>F	>8405	>9405

Step 1. The CATFIL listing shows SYSLD to start on track number $13B_{16}$.

CATALOG OF FILES FOR USER ID SYSTEM 1/ 1/ 74, 2 : 35: 40
DISC VOLUME # 1

FILE NAME	DISC ADDR	PASS WORD	INTEGRITY CODE	CURRENT FILE SIZE	MAX FILE SIZE	FILE TYPE	LRECL/ KEYLEN	PRECL WORDS
SMR	0114		7,4,4,7	0005	0005	RELREC	0040	0020
SMRWRK	0021		7,7,7,0	0002	000A	LINKSEQ		0080
SDQ	003E		4,4,4,4	0001	0001	RELREC	0040	0020
SYSLD	013B		4,4,4,4	0020	0020	RELREC	0040	0020
USRFTN	00EA		7,4,4,0	000E	0014	INDEXED	0008	0080
USRPLX	0009		7,4,4,0	0008	0014	INDEXED	0008	0080

Step 2. The starting sector address of SYSLD is calculated as:

$13B_{16}$ (Starting track no. of SYSLD).

$x58_{16}$ (Sectors/Track).

$6C48_{16}$ (Starting sector address of SYSLD).

Step 3. The MIP to modify is number C_{16}. Continue to step 4.

Step 4. The starting sector address of the MIP Directory is calculated as:

$6C48_{16}$ (Starting sector address of SYSLD).

+ 1 (Sector offset for MIP C_{16} Directory Words).

$6C49_{16}$ (Sector address for MIP Directory Words).

Step 5.

```
//DD 1 >6C49
6C49
    005F  0008  0211  0259  0028  0226  04CE  004F  ._......Y.(.&.N.O
    0250  012A  0015  025B  00DB  0010  0263  004B  .P.◆...[.[.....K
    0007  0267  0145  0017  0273  045B  0048  0299  .....E.....[.H..
    0087  0008  029F  008C  000B  02A5  0181  001B  ...........%....
```

Step 6. The directory words for MIP C_{16} are indicated by the box in step 5.

Step 7. The starting sector address of the MIP is calculated as:

$6C48_{16}$ (Starting sector address of SYSLD).

$+211_{16}$ (Starting sector address of MIP relative to SYSLD).

$6E59_{16}$ (Starting sector address of MIP C_{16}.

Step 8. The sector address to patch on the disc is calculated as:

$6E59_{16}$ (Starting sector address of MIP C_{16}.

$+ \ D_{16}$ (Offset in MIP from Sample excerpt from Table 2).

$6E66_{16}$ (Absolute disc sector address to be patched).

Step 9. The sector as it is currently on the disc is:

```
//DD 1 >6E66
6E66
    C8E2  CA21  C503  C514  0C0D  140F  0700  C8E2  H.J!E.E.......H.
    CA21  DF00  040A  1CE0  C536  8102  0700  8405  J!_.....E6......
    7C00  01D8  007A  007C  007B  007D  01D8  845C  ...X..........X.\
    0CFB  0700  C8E2  CA21  C503  C514  0CF3  1454  ....H.J!E.E....T
```

Step 10. Modify the sector with the patch. The modify command and a display of the sector after modification are shown below:

```
//MD 1 >6E66 >D >2102 >8102 >9405
//DD 1 >6E66
6E66
    C8E2  CA21  C503  C514  0C0D  140F  0700  C8E2  H.J!E.E.......H.
    CA21  DF00  040A  1CE0  C536  2102  8102  9405  J!_.....E6!.....
    7C00  01D8  007A  007C  007B  007D  01D8  845C  ...X..........X.\
    0CFB  0700  C8E2  CA21  C503  C514  0CF3  1454  ....H.J!E.E....T
```

Step 11. The patch is now on the disc. Repeat steps 2 through 10 for all the patches.

Table 1.

Standard Operating System Patch Guide

for the DS330 Disc

System Revision Letter: *B
Date: 18 Aug. 1975

Patch #	Description	Module Name	Absolute Sector	Word Offset into Sector	Present	Patch
1	Multitasking special case causes crash code >F	IOHEAD	>6D3	3	>0F01	>5400
				4	>8907	>02E4
2	User start job problem - go to READY but will not go to RUN	JUSTAR	>D44	>D	>0F01	>0F1A
				>E	>890A	>8901

Figure 1. Printout of Patches Applied to the Standard System

on a DS330 Disc

System Revision Letter: *B
Date: 18 Aug 1975

Patch # 1.

```
//DD 1 >6D3
06D3
    0F02  C0E1  8908  0F01  8907  1F07  C0E3  740A   ..∂.........∂...
    0502  C565  1900  C536  8105  C556  788D  1F72   ..E...E6..EV....
    07D3  0F68  15A9  B103  C516  C507  03E7  001F   .S...)1.E.E.....
    0002  000C  0001  074A  0000  0000  0000  0000   .......J........
//MD 1 >6D3 3 >5400 >02E4
//DD 1 >6D3
06D3
    0F02  C0E1  8908  5400  02E4  1F07  C0E3  740A   ..∂...T.....∂...
    0502  C565  1900  C536  8105  C556  788D  1F72   ..E...E6..EV....
    07D3  0F68  15A9  B103  C516  C507  03E7  001F   .S...)1.E.E.....
    0002  000C  0001  074A  0000  0000  0000  0000   .......J........
```

Patch # 2.

```
//DD 1 >D44
0D44
    8103  0700  C516  8115  C556  0103  C844  C8C4   ....E...EV..HDHD
    8102  0F01  8C0C  C0D1  890B  0F01  890A  1F0A   ......∂Q........
    C0D3  7408  0F00  8C03  B106  C516  C507  02E4   ∂S......1.E.E...
    1F72  0074  1C96  000F  0000  0000  0000  0000   ................
//MD 1 >D44 >D >0F1A >8901
//DD 1 >D44
0D44
    8103  0700  C516  8115  C556  0103  C844  C8C4   ....E...EV..HDHD
    8102  0F01  8C0C  C0D1  890B  0F1A  8901  1F0A   ......∂Q........
    C0D3  7408  0F00  8C03  B106  C516  C507  02E4   ∂S......1.E.E...
    1F72  0074  1C96  000F  0000  0000  0000  0000   ................
```

Table 2.

Nonstandard Operating System Patch Guide

System Revision Letter: *B
Date: 18 Aug. 1975

atch	Load Module	Load Mod Start Sector LS	MIP #	Directory Words Sector Offset DS	Directory Words Word Offset DW	Sector Offset Into MIP MS	Calc Abs Sector AS	Word Offset into Sector	Present	Patch
	SYSLD		0			>4B		3	>0F01	>5400
								4	>8907	>02E4
	SYSLD		>6C	>A	2	3		>D	>0F01	>0F1A
								>E	>890A	>8901

AS = LS + ({LS + DS} sector, DW word) + MS

AS = Absolute sector address of patch
LS = Absolute sector address of start of load module
DS = Directory words sector offset
DW = Directory words word offset
MS = Sector offset into MIP
( ) Signifies contents of particular word of particular sector (sector, word)

Table 1.

Standard Operating System Patch Guide

for the DS330 Disc

System Revision Letter: *B
Date: 29 Aug. 1975

Patch #	Description	Module Name	Absolute Sector	Word Offset into Sector	Present	Patch
3	Internal DMAC expansion system will not IPL with 2nd disc in system. Note: 2nd disc may not be added to Device Table until this patch is made. After patch made to system, re IPL and then add 2nd disc.	SYSTBL	>69E	4	>0000	>4000
4	DSCA overload condition - job submitted message may not be printed or may cause subsystem to error off with I/O attempted on unassigned LUN.	JSCOMT	>D28	7	>0F01	>0F02
				>B	>F800	>0808
				>C	>B10E	>8914
				>D	>C516	>F800
				>E	>C507	>782F

Figure 1. Printout of Patches Applied to the Standard System

on a DS330 Disc

System Revision Letter: *B
Date: 29 Aug 1975

Patch # 3.

```
//DD 1 >69E
069E
 0000  0000  0000  0000  0000  0000  0000  0000   ................
 0000  0000  0000  0000  0000  0000  0000  0000   ................
 0000  0000  0000  0000  0000  0000  0000  0000   ................
 0000  0000  0000  0000  0000  0000  0000  0000   ................
//MD 1 >69E 4 >4000
//DD 1 >69E
069E
 0000  0000  0000  0000  4000  0000  0000  0000   ........@.......
 0000  0000  0000  0000  0000  0000  0000  0000   ................
 0000  0000  0000  0000  0000  0000  0000  0000   ................
 0000  0000  0000  0000  0000  0000  0000  0000   ................
```

Patch # 4.

```
//DD 1 >D28
0D28
 8109  0700  840D  1706  C0E2  C521  8913  0F01   ........@.E!....
 8912  1F12  C0E3  F800  B10E  C516  C507  00A1   ....@...1.E.E..!
 02E4  1E1B  00A2  00A3  026A  00A4  00A5  2508   .....".#...$.%%.
 040F  2712  8102  0502  210A  2F01  8502  0102   ..'.....!./.....
//MD 1 >D28 7 >0F02
//MD 1 >D28 >B >0808 >8914 >F800 >782F
//DD 1 >D28
0D28
 8109  0700  840D  1706  C0E2  C521  8913  0F02   ........@.E!....
 8912  1F12  C0E3  0808  8914  F800  782F  00A1   ....@......../.!
 02E4  1E1B  00A2  00A3  026A  00A4  00A5  2508   .....".#...$.%%.
 040F  2712  8102  0502  210A  2F01  8502  0102   ..'.....!./.....
```

Table 2.

Nonstandard Operating System Patch Guide

System Revision Letter: *B
Date: 29 Aug. 1975

Load Module	Load Mod Start Sector LS	MIP #	Directory Words Sector Offset DS	Directory Words Word Offset DW	Sector Offset Into MIP MS	Calc Abs Sector AS	Word Offset into Sector	Present	Patch
SYSLD		0			>16		>4	>0000	>4000
(This is location 96 in Memory.)									
SYSLD		>68	9	>16	3		>7	>0F01	>0F02
							>B	>F800	>0808
							>C	>B10E	>8914
							>D	>C516	>F800
							>E	>C507	>782F

AS = LS + ({LS + DS} sector, DW word) + MS

AS = Absolute sector address of patch
LS = Absolute sector address of start of load module
DS = Directory words sector offset
DW = Directory words word offset
MS = Sector offset into MIP
( ) Signifies contents of particular word of particular sector (sector, word)

Table 1.

Standard Operating System Patch Guide

for the DS330 Disc

System Revision Letter: *B
Date: 14 Nov. 1975

Description	Module Name	Absolute Sector	Word Offset into Sector	Present	Patch
Job cancellation during submittal may result in multiple entry into the serially reuseable job management modules.	JINR	>767	>14	>F847	>F84A
User task waiting for time-of-day may not be awakened.	TSPOST	>783	>1F	>1A12	>1A13
		>784	>A	>7806	>780B
			>10	>7800	>CD20
			>11	>7804	>7805
			>12	>1D58	>7803
			>13	>1D63	>1D58
			>14	>1D5E	>1D63
			>15	>1D63	>1D5E
Concurrent users may enter the nonreentrant Create Task SVC.	SVCDRT	>7FD	3	>0200	>021E
Blocked relative record files on DS330 disc are not utilizing disc efficiency feature - Note: prior to making patch, all blocked relative record files on DS330 disc must be copied to an unblocked file (using DXCOPY); subsequent to making patch, copy the unblocked files back to blocked files.	FMPRIO	>8AB	>1B	>011A	>C881
			>1C	>991C	>7800
			>1D	>CA2F	>7800
			>1E	>C8E1	>7800
			>1F	>C862	>7800
		>8AC	0	>890D	>810D

Table 1.

Standard Operating System Patch Guide

for the DS330 Disc

System Revision Letter: *B
Date: 14 Nov. 1975

Description	Module Name	Absolute Sector	Word Offset into Sector	Present	Patch
Manual intervention with magnetic tape drive while accessing magnetic tape may result in hard >CE00 idle.	IDMT	>C16	>1C	>0089	>0081
Read/Write direct I/O to data terminal result in illegal op code error.	IDDT	>C5D	>D	>6F12	>6F14
Job cancellation during submittal may result in multiple entry into the serially reuseable job management modules.	OJCBPR	>D6C	3	>F847	>F84A
Ready jobs may be lost from system queue.	JREADY	>E2F	>D	>CD20	>7800

Figure 1. Printout of Patches Applied to the Standard System

on a DS330 Disc

System Revision Letter: *B
Date: 14 Nov. 1975

Patch # 5.

```
//DD 1 >767
0767
 7816  0736  8102  8907  0F04  C0E1  890F  0704  ...6......@.....
 242E  8110  0F02  C0E1  8911  0F07  C0E1  8912  $.....@.....@...
 0F04  890E  1F0E  C0E3  F847  1822  F805  0F01  ......@..G."....
 8D00  C565  1C1C  C536  0104  C556  2500  8101  ..E...E6..EV%...
//MD 1 >767 >14 >F84A
//DD 1 >767
0767
 7816  0736  8102  8907  0F04  C0E1  890F  0704  ...6......@.....
 242E  8110  0F02  C0E1  8911  0F07  C0E1  8912  $.....@.....@...
 0F04  890E  1F0E  C0E3  F84A  1822  F805  0F01  ......@..J."....
 8D00  C565  1C1C  C536  0104  C556  2500  8101  ..E...E6..EV%...
```

Patch # 6.

```
//DD 1 >783 >784
0783
 C516  0303  80DE  041F  C506  0301  80DB  0100  E....^..E....[..
 C8C1  C801  6F1D  CD20  780C  00D3  3C15  1C14  HAH...M ...S<...
 C603  CD20  7806  00CD  3C10  80CB  00CB  3C0D  F.M ...M<..K.K<.
 80C9  1CDA  C536  0100  C8C7  C84E  C502  1A12  .I.ZE6..HGHNE...
0784
 C537  1D69  1C8F  1C95  1E0F  1E10  00BB  60B9  E7...........;.9
 CD20  7C01  7806  1E09  00B5  60B3  CDC0  7CFB  M .......5.3M@..
 7800  7804  1D58  1D63  1D5E  1D63  78BF  1CEA  .....X...^...?..
 C536  0100  DB50  8100  0F00  8C11  0102  8410  E6.. [P..........
//MD 1 >783 >1F >1A13
//MD 1 >784 >A >780B
//MD 1 >784 >10 >CD20 >7805 >7803 >1D58 >1D63 >1D5E
//DD 1 >783 >784
0783
 C516  0303  80DE  041F  C506  0301  80DB  0100  E....^..E....[..
 C8C1  C801  6F1D  CD20  780C  00D3  3C15  1C14  HAH...M ...S<...
 C603  CD20  7806  00CD  3C10  80CB  00CB  3C0D  F.M ...M<..K.K<.
 80C9  1CDA  C536  0100  C8C7  C84E  C502  1A13  .I.ZE6..HGHNE...
0784
 C537  1D69  1C8F  1C95  1E0F  1E10  00BB  60B9  E7...........;.9
 CD20  7C01  780B  1E09  00B5  60B3  CDC0  7CFB  M .......5.3M@..
 CD20  7805  7803  1D58  1D63  1D5E  78BF  1CEA  M .....X...^.?..
 C536  0100  DB50  8100  0F00  8C11  0102  8410  E6.. [P..........
```

Figure 1. Printout of Patches Applied to the Standard System
on a DS330 Disc

System Revision Letter: *B
Date: 14 Nov. 1975

Patch # 7.

```
/DD 1 >7FD
7FD
 0C00  0000  2012  0200  0500  0000  3025  0A1F  .... .......0%..
 3200  0000  183F  1B20  2000  0000  180F  0F00  2....?.  .......
 3200  0000  2025  0C22  3200  0000  1834  1223  2... %."2....4.#
 3200  0000  1849  0D24  3200  0000  1839  8F00  2....I.$2....9..
/MD 1 >7FD 3 >021E
/DD 1 >7FD
7FD
 0C00  0000  2012  021E  0500  0000  3025  0A1F  .... .......0%..
 3200  0000  183F  1B20  2000  0000  180F  0F00  2....?.  .......
 3200  0000  2025  0C22  3200  0000  1834  1223  2... %."2....4.#
 3200  0000  1849  0D24  3200  0000  1839  8F00  2....I.$2....9..
```

Patch # 8.

```
/DD 1 >8AB >8AC
8AB
 C556  1101  C526  8107  C556  7809  C565  1901  EV..E&..EV..E...
 C536  0107  C556  8116  C536  5107  C556  1900  E6..EV..E6Q.EV..
 C536  0115  6F00  C556  CD20  7805  C536  0117  E6....EVM ..E6..
 C556  810D  7806  011A  991C  CA2F  C8E1  C862  EV......?..J/H.H.
8AC
 890D  010D  C82F  C565  1900  C536  5917  C556  ....H/E...E6Y.EV
 810E  0116  C82F  590E  810F  C536  0114  C556  ....H/Y...E6..EV
 991D  A128  010F  991A  B928  A111  011A  C885  ..!(....9(!...H.
 C536  6915  C556  CDA0  7818  011D  C82F  591A  E6..EVM ....H/Y.
/MD 1 >8AB >1B >C881 >7800 >7800 >7800 >7800
/MD 1 >8AC 0 >810D
/DD 1 >8AB >8AC
8AB
 C556  1101  C526  8107  C556  7809  C565  1901  EV..E&..EV..E...
 C536  0107  C556  8116  C536  5107  C556  1900  E6..EV..E6Q.EV..
 C536  0115  6F00  C556  CD20  7805  C536  0117  E6....EVM ..E6..
 C556  810D  7806  C881  7800  7800  7800  7800  EV....H.........
8AC
 810D  010D  C82F  C565  1900  C536  5917  C556  ....H/E...E6Y.EV
 810E  0116  C82F  590E  810F  C536  0114  C556  ....H/Y...E6..EV
 991D  A128  010F  991A  B928  A111  011A  C885  ..!(....9(!...H.
 C536  6915  C556  CDA0  7818  011D  C82F  591A  E6..EVM ....H/Y.
```

Figure 1. Printout of Patches Applied to the Standard System on a DS330 Disc

System Revision Letter: *B

Date: 14 Nov. 1975

Patch # 9.

```
//DD 1 >C16
0C16
    006D  9119  C7C7  0000  4800  4603  4408  4405  ....GG..H.F.D.D.
    C7B6  B502  C7B6  0115  C010  C7C7  2802  7D19  G65.G6..@.GG(...
    190A  C7B6  0502  C7B6  3800  7DFF  C990  0A01  ..G6..G68...I...
    C517  0000  0083  0081  0089  0085  0095  0000  E...............
//MD 1 >C16 >1C >0081
//DD 1 >C16
0C16
    006D  9119  C7C7  0000  4800  4603  4408  4405  ....GG..H.F.D.D.
    C7B6  B502  C7B6  0115  C010  C7C7  2802  7D19  G65.G6..@.GG(...
    190A  C7B6  0502  C7B6  3800  7DFF  C990  0A01  ..G6..G68...I...
    C517  0000  0083  0081  0081  0085  0095  0000  E...............
```

Patch # 10.

```
//DD 1 >C5D
0C5D
    C550  7400  1831  0035  0052  010B  811D  705E  EP...1.5.R.....^
    7060  C7C7  2001  3FFF  C502  6F12  CD80  781A  ..GG .?.E...M...
    0115  C7C7  3800  0201  C507  0083  002A  018A  ..GG8...E....◆..
    002A  002A  002A  002A  005C  005C  002F  002F  .◆.◆.◆.◆.\.\./..
//MD 1 >C5D >D >6F14
//DD 1 >C5D
0C5D
    C550  7400  1831  0035  0052  010B  811D  705E  EP...1.5.R.....^
    7060  C7C7  2001  3FFF  C502  6F14  CD80  781A  ..GG .?.E...M...
    0115  C7C7  3800  0201  C507  0083  002A  018A  ..GG8...E....◆..
    002A  002A  002A  002A  005C  005C  002F  002F  .◆.◆.◆.◆.\.\./..
```

Patch # 11.

```
//DD 1 >D6C
0D6C
    8958  1F58  C0D3  F847  0C3F  8959  1730  C0D2  .X.X@S.G.?.Y.0@R
    C520  815A  0702  8158  1F58  C0D3  7436  0F00  E .Z...X.X@S.6..
    8905  0709  C565  1904  C536  8101  C561  C556  ....E...E6..E.EV
    8959  0701  8158  1F58  C0D3  F800  0F09  C0E1  .Y...X.X@S....@.
//MD 1 >D6C 3 >F84A
//DD 1 >D6C
0D6C
    8958  1F58  C0D3  F84A  0C3F  8959  1730  C0D2  .X.X@S.J.?.Y.0@R
    C520  815A  0702  8158  1F58  C0D3  7436  0F00  E .Z...X.X@S.6..
    8905  0709  C565  1904  C536  8101  C561  C556  ....E...E6..E.EV
    8959  0701  8158  1F58  C0D3  F800  0F09  C0E1  .Y...X.X@S....@.
```

Figure 1. Printout of Patches Applied to the Standard System

on a DS330 Disc

System Revision Letter: *B
Date: 14 Nov. 1975

Patch # 12.

```
//DD 1 >E2F
0E2F
    00AB  7417  0501  8102  07FF  6103  CDA0  7C12   .+.........M ..
    000D  2F10  8501  0700  6103  CD20  780C  0700   ../.......M ....
    8405  781A  00A6  0042  01CF  01D0  0268  00A7   .....&.B.D.P...
    1A67  24FB  006E  0102  C565  1D01  C536  6110   ..$.....E...E6..
//MD 1 >E2F >D >7800
//DD 1 >E2F
0E2F
    00AB  7417  0501  8102  07FF  6103  CDA0  7C12   .+.........M ..
    000D  2F10  8501  0700  6103  7800  780C  0700   ../.............
    8405  781A  00A6  0042  01CF  01D0  0268  00A7   .....&.B.D.P...
    1A67  24FB  006E  0102  C565  1D01  C536  6110   ..$.....E...E6..
```

Table 2.

Nonstandard Operating System Patch Guide

System Revision Letter: *B

Date: 14 Nov. 1975

Patch #	Load Module	Load Mod Start Sector LS	MIP #	Directory Words Sector Offset DS	Directory Words Word Offset DW	Sector Offset Into MIP MS	Calc Abs Sector AS	Word Offset Into Sector	Present	Patch
5	SYSLD		0			>F		>14	>F847	>F84A
6	SYSLD		0			>FB		>1F	>1A12	>1A13
						>FC		>A	>7806	>780B
								>10	>7800	>CD20
								>11	>7804	>7805
								>12	>1D58	>7803
								>13	>1D63	>1D58
								>14	>1D5E	>1D63
								>15	>1D63	>1D5E
	SYSLD		0			>175		3	>0200	>021E
8	SYSLD		>A	0	>1C	2		>1B	>011A	>C881
								>1C	>991C	>7800
								>1D	>CA2F	>7800
								>1E	>C8E1	>7800
								>1F	>C862	>7800
						3		0	>890D	>810D
	SYSLD		>4F	7	>B	6		>1C	>0089	>0081
	SYSLD		>58	8	6	0		>D	>6F12	>6F14
	SYSLD		>6F	>A	>B	>1A		3	>F847	>F84A
	SYSLD		>7F	>B	>1B	2		>D	>CD20	>7800